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[22TCSE215]

FIFTH SEMESTER B.Tech EXAMINATION, SEPTEMBER-2024

CAUSAL REASONING

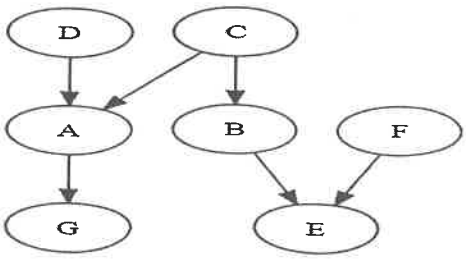
Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	What is inductive reasoning? Provide an example.	L1	CO1	**02**
	b.	How does the Fundamental Problem of Causal Inference relate to counterfactuals?	L1	CO2	**02**
	c.	Define a causal Bayesian network.	L1	CO3	**02**
	d.	Why is covariate adjustment important in causal inference?	L1	CO4	**02**
	e.	What is the primary goal of A/B testing in social media experiments?	L1	CO4	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain inductive reasoning and discuss its strengths and limitations. Provide an example to support your explanation.	L2	CO1	**05**
		OR			
	b.	Define factuals and counterfactuals in the context of causal reasoning. How are they used in understanding causality, and what are the challenges associated with counterfactual reasoning?	L1	CO1	**05**
3.	**a.**	Describe the process of estimating the Average Treatment Effect (ATE) using the Potential Outcomes framework.	L1	CO2	**05**
		OR			
	b.	Discuss the concept of Unconfoundedness and its importance in the context of causal inference.	L6	CO2	**05**
4.	**a.**	How do researchers distinguish between correlation and causation? Discuss the potential pitfalls of assuming causation based on correlation with an example.	L1	CO1	**05**
		OR			
	b.	Explain the d- separation with your own example.	L2	CO4	**05**

5.	**a.**	Explain how Bayesian inference can be applied to estimate causal effects in observational studies.	L2	CO3	**05**
		OR			
	b.	How does association makes causation in causal reasoning? Justify with an example related to Bayesian network.	L1	CO3	**05**
6.	**a.**	Describe the impact of omitting an important covariate from a regression model.	L1	CO4	**05**
		OR			
	b.	Explain the relationship between covariate balance and the internal validity of an experiment.	L2	CO4	**05**
7.	**a.**	Explain an application of causal reasoning in healthcare and how it improves decision-making.	L2	CO4	**05**
		OR			
	b.	Describe the process of conducting an A/B test in an e-commerce setting, including the steps involved.	L1	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Compare and contrast inductive, deductive, and abductive reasoning. Discuss the strengths and limitations of each method, and explain how they can be applied in different fields such as science, law, and medicine. Provide examples to illustrate how these reasoning methods lead to different types of conclusions.	L4	CO1	**10**
		OR			
	b.	Differentiate between association and correlation, and discuss their roles in statistical analysis. How can researchers ensure that correlations are not mistaken for causation? provide examples.	L5	CO1	**10**
9.	**a.**	Explain the Fundamental Problem of Causal Inference and its implications for causal research. Discuss how this problem affects the estimation of both Individual Treatment Effects (ITE) and Average Treatment Effects (ATE).	L2	CO2	**10**
		OR			

	b.	Discuss the relationship between the Fundamental Problem of Causal Inference and the assumption of Conditional Exchangeability. How does Conditional Exchangeability help in addressing the challenges posed by the Fundamental Problem?	L6	CO2	**10**
10.	**a.**	Explain Simpson's Paradox and its implications for data analysis. How can this paradox lead to misleading conclusions in research and decision-making? Discuss strategies to detect and avoid Simpson's Paradox in statistical analysis, using real-world examples to illustrate your points.	L5	CO1	**10**
		OR			
	b.	A social media platform wants to test a new algorithm that prioritizes content based on user engagement rather than chronological order. The platform plans to conduct an A/B test where one group of users (Group A) will see content in chronological order, while another group (Group B) will see content prioritized by engagement. As a data scientist, describe how you would design this A/B test to balance the platform's goals with the ethical implications. What steps would you take to measure and mitigate potential negative effects?	L6	CO4	**10**
11.	**a.**	Compare and contrast the impact of sensitivity, specificity, and prevalence in the context of COVID-19 testing. How do these factors influence public health decisions related to mass testing and screening programs?	L5	CO3	**10**
		OR			
	b.	Try to figure out some independence and dependence relationship in the following causal graph. 1. Is $D \perp C$? 2. Is $D \perp C \mid A$? 3. Is $D \perp C \mid G$? 4. Is $A \perp F$? 5. Is $A \perp F \mid E$? 6. Is $A \perp F \mid E, C$?  	L5	CO3	**10**

12.	**a.**	Discuss the role of covariate adjustment in distinguishing between causation and correlation in observational studies.	L6	CO4	**10**
		OR			
	b.	Critically analyze how the concepts of no backdoor paths and covariate balance work together to ensure unbiased causal inference in randomized controlled trials (RCTs).	L4	CO4	**10**
13.	**a.**	Discuss the ethical issues associated with A/B testing in e-commerce and social media, and how these issues can impact policy setting. Provide examples to support your discussion.	L6	CO4	**10**
		OR			
	b.	Evaluate the ethical considerations of using A/B testing in designing algorithms that influence content visibility on social media platforms. How can these considerations inform policy development?	L5	CO4	**10**